

Solutions to CMI Mock Test (Part A)

Q1.

Correct Options: C, D

By the Polynomial Identity Theorem, if two polynomials of degree $\leq n$ agree at $n + 1$ distinct points, they are identical everywhere. All given expressions have maximum possible degree 2. We test each at the three distinct points $x = a$, $x = b$, and $x = c$.

Expression A evaluates to 1 at a , b , and c . The constant polynomial $P(x) = 1$ has degree ≤ 2 and agrees at 3 points, so Expression A is identically 1. Its degree is 0.

Expression B is just x multiplied to expression A, hence it is identically x . Its degree is 1.

Expression C evaluates to a^2 , b^2 , and c^2 . The polynomial $P(x) = x^2$ has degree ≤ 2 and agrees at 3 points, so Expression C is identically x^2 . Its degree is exactly 2.

Expression D evaluates to a^3 , b^3 , and c^3 . We construct a degree ≤ 2 polynomial mimicking x^3 at these roots: $P(x) = x^3 - (x - a)(x - b)(x - c)$. Expanding this cancels the x^3 term, leaving $P(x) = (a + b + c)x^2 - (ab + bc + ca)x + abc$. Assuming $a + b + c \neq 0$, the leading coefficient is non-zero, making its degree exactly 2.

Q2.

Correct Options: A, C, D

An idempotent matrix satisfies $A^2 = A$. To test the invertibility of $I + cA$, we try to construct its inverse in the form $I + dA$. Multiplying:

$$(I + cA)(I + dA) = I + dA + cA + cdA^2 = I + (c + d + cd)A.$$

For this to equal I , we need $c + d + cd = 0$, giving $d = -\frac{c}{c+1}$. Thus $I + cA$ is invertible whenever $c + 1 \neq 0$, i.e. $c \neq -1$.

Options A ($c = 1$), C ($c = 2$), and D ($c = 1/2$) are invertible. Option B ($c = -1$) is not.

Q3.

Correct Options: A, B, D

A: Since $0 \leq \frac{x^n}{1+x} \leq x^n$ on $[0, 1]$, integrating gives $0 \leq I_n \leq \frac{1}{n+1}$. By the Squeeze Theorem, $\lim_{n \rightarrow \infty} I_n = 0$. **True.**

B: Adding consecutive terms simplifies the integrand:

$$I_n + I_{n+1} = \int_0^1 \frac{x^n(1+x)}{1+x} dx = \int_0^1 x^n dx = \frac{1}{n+1}.$$

True.

C & D: Because $x^n > x^{n+1}$ on $(0, 1)$, it follows strictly that $I_n > I_{n+1}$. Replacing I_{n+1} with I_n in the identity from (B) gives $I_n + I_n > \frac{1}{n+1}$, so $I_n > \frac{1}{2(n+1)}$. Thus **D is True** and **C is False**.

Q4.

Correct Options: A, C, D

A: Since $f(1/n) = 0$ and $1/n \rightarrow 0$, continuity forces $f(0) = 0$. Between any two consecutive roots $1/n$ and $1/(n+1)$, Rolle's Theorem guarantees a root of $f'(x)$. These roots also accumulate at 0, forcing $f'(0) = 0$. By induction, $f^{(k)}(0) = 0$ for all k . **True.**

B: False. Flat functions (such as $e^{-1/x^2} \sin(\pi/x)$ properly extended at the origin) satisfy this condition without being identically zero.

C: Because every derivative at 0 is zero, the Maclaurin series is $0 + 0x + 0x^2 + \dots = 0$. This trivially converges to 0 for all real x . **True.**

D: We have roots of $f(x)$ at $x = 1/2, 1/3, 1/4$. Rolle's Theorem on $f(x)$ gives roots of $f'(x)$ in $(1/3, 1/2)$ and $(1/4, 1/3)$. Applying Rolle's again on $f'(x)$ between those two new roots guarantees some c strictly between $1/4$ and $1/2$ with $f''(c) = 0$. **True.**

Q5.

Correct Options: A, D

Taking the natural logarithm converts the infinite product into a Riemann-like sum:

$$\ln(a_n) = \sum_{k=1}^n \ln\left(1 + \frac{k}{n^2}\right).$$

Using the first-order expansion $\ln(1+x) \approx x$ for small x , this becomes

$$\ln(a_n) \approx \sum_{k=1}^n \frac{k}{n^2} = \frac{1}{n^2} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2n}.$$

As $n \rightarrow \infty$, $\frac{n+1}{2n} \rightarrow \frac{1}{2}$. Therefore $\lim_{n \rightarrow \infty} \ln(a_n) = 1/2$. (**D is True.**)

Exponentiating gives $\lim_{n \rightarrow \infty} a_n = e^{1/2} = \sqrt{e}$. (**A is True, B is False.**)

Testing early terms: $a_1 = 2$ and $a_2 = (1 + 1/4)(1 + 2/4) = 1.875$. The sequence decreases, so it is not strictly increasing. (**C is False.**)

Q6.

Correct Options: A, B

A: If a function is uniformly continuous on $[0, \infty)$ and its improper integral converges, the function must decay to 0 as $x \rightarrow \infty$. **True.**

B: A uniformly continuous function cannot grow faster than a linear rate. By definition, for $\varepsilon = 1$ there exists δ bounding the step size. Using the triangle inequality $\lfloor x/\delta \rfloor$ times shows $|f(x)|$ is bounded by a line $Ax + B$. **True.**

C & D: False. The function $f(x) = x$ is uniformly continuous but unbounded (disproving C). The function $f(x) = \sin(x^3)/x$ can be smoothed at 0 to be uniformly continuous, but its derivative oscillates wildly to infinity (disproving D).

Q7.

Correct Options: A, B, C, D

A & B: By Fermat's Theorem on sums of two squares, an odd prime p is a sum of two integer squares if and only if $p \equiv 1 \pmod{4}$. This base representation is unique up to order and sign. **(Both True.)**

C: For $p = 13$, the unique base pair is $\{2, 3\}$. Accounting for coordinate order (x, y) versus (y, x) and signs $(\pm, \pm)$, there are $2 \times 4 = 8$ distinct integer solutions. **True.**

D: If p is an odd prime, the sum $x^2 + y^2$ is odd. This requires exactly one square to be odd and the other even, so one base number is odd and the other is even. Their sum $x + y$ must therefore be odd. **True.**

Q8.

Correct Options: A, B, C, D

For A, B, and C, each of the n elements of S has exactly 3 mutually exclusive placement choices:

A: Placed in A only, B only, or both (to satisfy $A \cup B = S$). This gives 3^n ways.

B: Placed in neither, B only, or both (to satisfy $A \subseteq B$). This gives 3^n ways.

C: Placed in A only, B only, or neither (to satisfy $A \cap B = \emptyset$). This gives 3^n ways.

D: Choosing A and B to both have size k gives $\binom{n}{k}^2$ ways. Summing over all k from 0 to n gives $\sum_{k=0}^n \binom{n}{k}^2$, which by Vandermonde's Identity equals $\binom{2n}{n}$.

(All True.)

Q9.

Correct Options: A, B, C

A & B: Since D lies on the circumcircle and bisects arc BC , angle chasing shows $\triangle BDI$ and $\triangle CDI$ are isosceles, giving $DB = DC = DI$. Since D is equidistant from all three vertices of $\triangle BIC$, it is the circumcentre of $\triangle BIC$. **(Both True.)**

C: By the intersecting chords theorem, the power of I with respect to the circumcircle is $AI \cdot ID$. The standard distance formula from the incentre to the circumcentre makes this power $R^2 - d^2$. **True.**

D: False. The triangle $\triangle BIC$ is obtuse (since $\angle BIC = 90^\circ + A/2 > 90^\circ$), and the orthocentre of an obtuse triangle lies outside it. It cannot coincide with the circumcentre D .

Q10.

Correct Options: A, B, C, D

B: Expanding $\frac{\sin((2n+1)\theta)}{\cos^{2n+1}\theta}$ via De Moivre's formula produces a polynomial $P(x)$ of degree n whose roots x_k are exactly $\tan^2(\frac{k\pi}{2n+1})$. **True.**

A & D: The constructed polynomial is

$$P(x) = \binom{2n+1}{1} - \binom{2n+1}{3}x + \cdots + (-1)^n \binom{2n+1}{2n+1}x^n.$$

By Vieta's formulas, the sum of roots is $S_n = \binom{2n+1}{2n-1} / \binom{2n+1}{2n+1} = n(2n+1)$. (**A is True.**) The product of roots is the constant term divided by the leading coefficient's magnitude, giving $\binom{2n+1}{1} / 1 = 2n+1$. (**D is True.**)

C: Using $\sec^2 \theta = 1 + \tan^2 \theta$, the required sum transforms to $\sum 1 + S_n = n + n(2n+1) = 2n^2 + 2n = 2n(n+1)$. **True.**